

Stat 517
Homework # 6

1. Prove that $\bar{X}$, the mean of a random sample of size n from a distribution that is $N(\theta, \sigma^2)$, $-\infty < \theta < \infty$, is for every known $\sigma^2 > 0$, an efficient estimator of θ .
2. Let X have a gamma distribution with $\alpha = 4$ and $\beta = \theta > 0$.
 - (a) Find the Fisher information $I(\theta)$.
 - (b) If $X_1, \dots, X_n$ is a random sample from this distribution, show that the MLE of θ is an efficient estimator of θ .
3. Let X be $N(0, \theta)$, $0 < \theta < \infty$.
 - (a) Find the Fisher information $I(\theta)$.
 - (b) If $X_1, X_2, \dots, X_n$ is a random sample from this distribution, show that the mle of θ is an efficient estimator of θ .
 - (c) What is the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta)$?
4. If $X_1, X_2, \dots, X_n$ is a random sample from a distribution with pdf

$$f(x; \theta) = \begin{cases} \frac{3\theta^3}{(x + \theta)^4} & 0 < x < \infty, 0 < \theta < \infty \\ 0 & \text{elsewhere,} \end{cases}$$

show that $Y = 2\bar{X}$ is an unbiased estimator of θ and determine its efficiency.

5. Let $X_1, \dots, X_n$ be a random sample from a distribution with pmf $p(x; \theta) = \theta^x (1 - \theta)^{1-x}$, $x = 0, 1$, where $0 < \theta < 1$. We wish to test $H_0 : \theta = 1/3$ versus $H_1 : \theta \neq 1/3$.
 - (a) Find Λ and $-2 \log \Lambda$.
 - (b) Determine the Wald-type test.
 - (c) What is the Rao's score statistic?
6. Let $X_1, \dots, X_n$ be a random sample from a $\Gamma(\alpha, \beta)$ distribution where α is known and $\beta > 0$. The $\Gamma(\alpha, \beta)$ pdf is

$$f(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

Determine the likelihood ratio test for $H_0 : \beta = \beta_0$ against $H_1 : \beta \neq \beta_0$.

7. Let $X_1, \dots, X_n$ be iid, each with the distribution having pdf $f(x; \theta_1, \theta_2) = (1/\theta_2) e^{-(x-\theta_1)/\theta_2}$ where $\theta_1 \leq x < \infty$, $-\infty < \theta_2 < \infty$, zero elsewhere. Find the maximum likelihood estimators of θ_1 and θ_2 .
8. Let $X_1, \dots, X_n$ be a random sample from the distribution $N(\theta_1, \theta_2)$. Show that the likelihood ratio principle for testing $H_0 : \theta_2 = \theta'_2$ specified, and θ_1 unspecified against $H_1 : \theta_2 \neq \theta'_2$, θ_1 unspecified, leads to a test that rejects when $\sum_i (x_i - \bar{x})^2 \leq c_1$ or $\sum_i (x_i - \bar{x})^2 \geq c_2$, where $c_1 < c_2$ are selected appropriately.